Light

GCSE Computer Science — CPU Architecture (Set 2)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Cpu Architecture

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which of the following is NOT a component found inside the CPU?

- **A.** Hard disk drive
- **B.** Arithmetic Logic Unit
- **C.** Control Unit
- **D.** Cache memory

Q2 (1 mark)

A program needs to calculate 150 minus 37.
Which CPU component will perform this subtraction?

- **A.** The Memory Address Register

- B. The Arithmetic Logic Unit
- C. The Program Counter
- D. The Control Unit

Q3 (1 mark)

What does the Control Unit send to other components to coordinate their operations?

- A. Control signals
- B. Data values for calculation
- C. Memory addresses
- D. Program source code

Q4 (1 mark)

Why is cache memory faster to access than main memory (RAM)?

- A. Cache is located closer to the CPU and uses faster technology
- B. Cache has a larger storage capacity
- C. Cache is connected via USB
- D. Cache stores data permanently

Q5 (1 mark)

Which type of memory has the fastest access time?

- ○ **A.** Secondary storage (SSD)
- ○ **B.** CPU registers
- ○ **C.** Main memory (RAM)
- ○ **D.** Cache memory

Q6 (1 mark)

In Von Neumann architecture, how are instructions processed?

- ○ **A.** Instructions are executed one at a time in sequence
- ○ **B.** Multiple instructions are executed simultaneously
- ○ **C.** Instructions are executed in random order
- ○ **D.** All instructions are executed before any data is processed

Q7 (1 mark)

The Memory Address Register (MAR) is connected to which bus?

- ○ **A.** The address bus
- ○ **B.** The data bus
- ○ **C.** The control bus
- ○ **D.** The USB bus

Q8 (1 mark)

Data can flow in both directions through the Memory Data Register (MDR).
What does this mean?

- ○ **A.** The MDR can only send data to memory
- ○ **B.** The MDR can both send data to memory and receive data from memory
- ○ **C.** The MDR can only receive data from memory
- ○ **D.** The MDR converts data between binary and denary

Q9 (1 mark)

After an instruction is fetched, what normally happens to the Program Counter?

- ○ **A.** It is reset to zero
- ○ **B.** It is incremented to point to the next instruction
- ○ **C.** It is cleared and the program ends
- ○ **D.** It remains unchanged

Q10 (1 mark)

What is the relationship between the Accumulator and the ALU?

- ○ **A.** The Accumulator stores results produced by the ALU
- ○ **B.** The Accumulator controls when the ALU operates

- ○ **C.** The Accumulator and ALU are the same component
- ○ **D.** The ALU stores data for the Accumulator to process

Q11 (1 mark)

Clock speed is measured in Hertz (Hz).
What does one Hertz represent?

- ○ **A.** One byte of data
- ○ **B.** One cycle per second
- ○ **C.** One instruction completed
- ○ **D.** One core in the processor

Q12 (1 mark)

A computer has a CPU with 8 cores.
What does this specification mean?

- ○ **A.** The CPU has 8 independent processing units
- ○ **B.** The CPU can store 8 gigabytes of data
- ○ **C.** The CPU runs at 8 GHz
- ○ **D.** The CPU has 8 megabytes of cache

Q13 (2 marks)

During the fetch stage of the fetch-decode-execute cycle, several operations occur.

Which sequence correctly describes what happens during the fetch stage?

- ○ **A.** The instruction is executed, then the result is stored in memory
- ○ **B.** The ALU performs a calculation, then the Control Unit checks the result
- ○ **C.** The address in the PC is copied to the MAR, memory is accessed, and the instruction is loaded into the MDR
- ○ **D.** The Control Unit decodes the instruction and determines what operation to perform

Q14 (2 marks)

What is the main purpose of the decode stage in the fetch-decode-execute cycle?

- ○ **A.** To convert binary instructions into electrical signals and determine what operation is required
- ○ **B.** To fetch the next instruction from memory
- ○ **C.** To store the result of the calculation
- ○ **D.** To increment the Program Counter

Q15 (2 marks)

Cache memory improves performance based on the principle of 'locality of reference'.
What does this principle mean?

- ○ **A.** Programs always execute instructions in reverse order
- ○ **B.** Programs tend to access the same data or nearby data

repeatedly over short periods of time

- ○ **C.** Data stored in cache is never needed more than once
- ○ **D.** The CPU only accesses memory at the start of a program

Q16 (2 marks)

CPUs contain both general-purpose registers and special-purpose registers.
Which statement correctly describes the difference?

- ○ **A.** General-purpose registers can only store addresses, special-purpose can store any data
- ○ **B.** General-purpose registers can store various types of data for calculations, while special-purpose registers have specific functions like the Program Counter
- ○ **C.** There is no difference; all registers work the same way
- ○ **D.** Special-purpose registers are faster than general-purpose registers

Q17 (2 marks)

A user 'overclocks' their CPU by increasing the clock speed beyond the manufacturer's specification.
What are the potential consequences of this action?

- ○ **A.** The CPU will use less power and run cooler
- ○ **B.** The CPU will have more cores available
- ○ **C.** The CPU's cache size will automatically increase
- ○ **D.** The CPU may perform faster but generates more heat and may

become unstable

Q18 (2 marks)

When the ALU compares two values, how is the result of the comparison typically indicated?

- ○ **A.** The result is displayed on the monitor
- ○ **B.** The compared values are deleted from memory
- ○ **C.** Status flags or condition codes are set to indicate the outcome
- ○ **D.** The clock speed automatically adjusts

Q19 (2 marks)

The 'stored program concept' is a key feature of Von Neumann architecture.
What does this concept mean?

- ○ **A.** Programs must be stored on external USB drives
- ○ **B.** Programs can only be stored in read-only memory
- ○ **C.** Programs are stored in the same memory as data and can be modified like data
- ○ **D.** Each program requires its own dedicated memory chip

Q20 (2 marks)

A software application is described as 'multi-threaded'.
How does this relate to CPU cores?

- ○ **A.** Multi-threaded applications can only run on single-core processors
- ○ **B.** Multi-threaded applications disable all cores except one
- ○ **C.** Multi-threaded applications require exactly two cores to function
- ○ **D.** Multi-threaded applications can distribute work across multiple cores, potentially using them simultaneously

Q21 (2 marks)

When the CPU modifies data that is stored in cache, the corresponding data in main memory becomes outdated.
What is this situation called?

- ○ **A.** Cache hit
- ○ **B.** Cache miss
- ○ **C.** Cache coherence problem
- ○ **D.** Cache overflow

Q22 (2 marks)

A student wants to buy a new computer for video editing.
Which combination of CPU specifications would be most important to consider?

- ○ **A.** Only the brand name of the processor
- ○ **B.** Only the physical size of the CPU
- ○ **C.** Only the colour of the computer case

○ **D.** Clock speed, number of cores, and cache size all contribute to performance

Q23 (2 marks)

During a memory write operation, describe how the MAR and MDR work together.

○ **A.** The MDR sends an address to the MAR, which stores it in memory

○ **B.** Both registers hold the same data and write simultaneously

○ **C.** The MAR holds the destination address while the MDR holds the data to be written to that address

○ **D.** The MAR performs the write operation while the MDR controls timing

Q24 (3 marks)

Describe the complete fetch-decode-execute cycle for a simple ADD instruction that adds two numbers from memory.
Which description most accurately represents this process?

○ **A.** The ADD instruction is typed by the user, the CPU displays the result, and the program ends

○ **B.** The ALU decides which instruction to execute, fetches it from cache, and stores the result on the hard drive

○ **C.** All three stages happen simultaneously to save time, with results going directly to the monitor

○ **D.** Fetch: PC address copied to MAR, instruction retrieved to MDR, PC incremented. Decode: Control Unit interprets the ADD instruction

and identifies operand addresses. Execute: Data fetched from memory, ALU performs addition, result stored in Accumulator

Q25 (3 marks)

A computer has a very fast CPU (4.5 GHz, 8 cores, 16MB cache), but a user complains that loading large spreadsheets is slow.
Analyse why the fast CPU might not solve this problem.

- ○ **A.** The CPU is defective because fast CPUs should never have slow loading times
- ○ **B.** The CPU cores are incompatible with spreadsheet software
- ○ **C.** The cache is too large and slows down the processor
- ○ **D.** Loading files from storage is limited by storage speed, not CPU speed — the CPU must wait for data to arrive from the slower hard drive or SSD before it can process it

Q26 (3 marks)

An embedded system in a washing machine has a CPU with a much lower clock speed (50 MHz) than a desktop computer (3 GHz).
Evaluate why this lower specification is appropriate for this application.

- ○ **A.** The washing machine manufacturer couldn't afford a better CPU
- ○ **B.** Lower specifications are never appropriate; faster is always better
- ○ **C.** The washing machine needs the lower clock speed to spin at the correct RPM

○ **D.** The washing machine's tasks are simple and predictable, require low power consumption, produce less heat, and don't need the processing power of a desktop CPU

Q27 (3 marks)

Modern CPUs use a technique called 'pipelining' to improve performance.
Analyse how pipelining relates to the fetch-decode-execute cycle.

○ **A.** Pipelining replaces the fetch-decode-execute cycle with a completely different process

○ **B.** Pipelining stores all instructions in cache before executing any of them

○ **C.** Pipelining allows different stages of different instructions to occur simultaneously — while one instruction executes, another decodes, and another fetches — increasing instruction throughput

○ **D.** Pipelining eliminates the need for the decode stage entirely

Q28 (3 marks)

When cache memory is full and new data needs to be loaded, existing data must be removed.
Analyse the considerations involved in deciding which data to remove from cache.

○ **A.** The oldest data is always removed regardless of how often it's used

○ **B.** All cache data is deleted and the cache starts fresh each time

○ **C.** Cache replacement policies consider factors like how recently or frequently data was accessed, aiming to keep data most likely to be needed soon while removing data least likely to be needed

○ **D.** Data is removed randomly with no consideration of usage patterns

Q29 (3 marks)

A programmer finds that their application runs at the same speed on both a 4-core and an 8-core processor.
Analyse why doubling the cores did not improve performance.

○ **A.** The 8-core processor must be defective

○ **B.** Having more cores always slows down applications

○ **C.** The programmer must have disabled the extra cores accidentally

○ **D.** The application is likely single-threaded and can only utilise one core at a time — additional cores sit idle because the software cannot distribute work across them

Q30 (3 marks)

CPU architectures are sometimes classified as RISC (Reduced Instruction Set Computing) or CISC (Complex Instruction Set Computing).
Analyse how these approaches differ in their design philosophy.

○ **A.** RISC and CISC are just different brand names for the same technology

○ **B.** RISC processors cannot perform complex calculations, while CISC processors cannot perform simple calculations

○ **C.** RISC uses simple instructions that execute quickly in few cycles, relying on software to combine them for complex tasks. CISC uses complex instructions that do more work per instruction but may take more cycles to execute

○ **D.** RISC is obsolete and all modern processors use only CISC

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.